

$$1. (a) \quad y(kT) = \sum_{h=0}^k x(hT) = \sum_{h=0}^{k-1} x(hT) + x(kT) = y((k-1)T) + x(kT)$$

$$Y(z) = z^{-1}Y(z) + Z[x(kT)] = z^{-1}Y(z) + \frac{Tz^{-1}}{(1-z^{-1})^2}$$

$$\therefore Y(z) = \frac{Tz^{-1}}{(1-z^{-1})^3}, \quad y(0) = \lim_{z \rightarrow \infty} Y(z) = 0.$$

$$(b) \quad \frac{X(z)}{z} = \frac{z}{z(z-1)(z^2-\sqrt{2}z+1)}$$

$$= \frac{A}{z} + \frac{B}{z-1} + \frac{Cz+D}{z^2-\sqrt{2}z+1} \quad \cos 45^\circ = \frac{\sqrt{2}}{2} \quad (z^2-8)$$

$$A(z-1)(z^2-\sqrt{2}z+1) + Bz(z^2-\sqrt{2}z+1) + (Cz+D)z(z-1) = z$$

$$A(z^3-\sqrt{2}z^2+z-z^2+\sqrt{2}z-1) + B(z^3-\sqrt{2}z^2+z) + C(z^3-Cz^2+Dz^2-Dz) = z$$

$$Az^3 + A(-\sqrt{2}-1)z^2 + A(1+\sqrt{2})z - A + Bz^3 - \sqrt{2}Bz^2 + Bz + Cz^3 - Cz^2 + Dz^2 - Dz = z$$

$$(A+B+C)z^3 + (-\sqrt{2}A-A-\sqrt{2}B-C+D)z^2 + (A+\sqrt{2}A+B-D)z + (-A) = z$$

$$\Rightarrow A=0, \quad (B+C)z^3 + (-\sqrt{2}B-C+D)z^2 + (B-D)z = z$$

$$B+C=0, \quad -\sqrt{2}B-C+D=0, \quad B-D=1, \quad B=D+1$$

$$D+C+1=0, \quad -\sqrt{2}D-\sqrt{2}-C+D=0$$

$$C=-1-D, \quad (\sqrt{2}-1)D = -\sqrt{2}-C = -\sqrt{2}+1+D$$

$$(\sqrt{2}-2)D = -\sqrt{2}+1$$

$$D = \frac{1-\sqrt{2}}{\sqrt{2}-2}$$

$$\Rightarrow A=0, \quad B=1+\frac{\sqrt{2}}{2}, \quad C=-1-\frac{\sqrt{2}}{2}, \quad D=\frac{\sqrt{2}}{2}$$

$$\Rightarrow \frac{X(z)}{z} = \frac{1+\frac{\sqrt{2}}{2}}{z-1} + \frac{-(1+\frac{\sqrt{2}}{2})z+\frac{\sqrt{2}}{2}}{z^2-\sqrt{2}z+1}$$

$$= \frac{1+\frac{\sqrt{2}}{2}z^{-1}}{1-z^{-1}} + \frac{-z^{-1}-\frac{\sqrt{2}}{2}z^{-1}+\frac{\sqrt{2}}{2}z^{-2}}{1-\sqrt{2}z^{-1}+z^{-2}}$$

$$\begin{aligned}
 2. (a) \quad E(s) &= R(s) - H(s)C(s) = R(s) - C(s), \quad E^*(s) = R^*(s) - C^*(s) \\
 C(s) &= G_1(s)G_2(s)E^*(s), \quad C^*(s) = [G_1G_2(s)]^+ E^*(s) \\
 \Rightarrow C^*(s) &= [G_1G_2(s)]^+ R^*(s) - [G_1G_2(s)]^+ C^*(s) \\
 \Rightarrow C(z) &= \frac{G_1G_2(z) \cdot R(z)}{1 + G_1G_2(z)}
 \end{aligned}$$

$$\begin{aligned}
 G_1G_2(z) &= \mathcal{Z}[G_1(s) \cdot G_2(s)] = (1-z^{-1}) \mathcal{Z}\left[\frac{G_2(s)}{s}\right] = (1-z^{-1}) \mathcal{Z}\left[\frac{\alpha}{s(s+1)}\right] \\
 &= \alpha(1-z^{-1}) \frac{(1-e^{-T})z^{-1}}{(1-z^{-1})(1-e^{-T}z^{-1})} \\
 &= \frac{0.0952\alpha z^{-1}}{1 - 0.9048z^{-1}}
 \end{aligned}$$

$$R(z) = \frac{1}{1-z^{-1}}, \quad 1 + G_1G_2(z) = \frac{1 - 0.9048z^{-1} + 0.0952\alpha z^{-1}}{1 - 0.9048z^{-1}}$$

$$\Rightarrow C(z) = \frac{0.0952\alpha z^{-1}}{(1-z^{-1})(1 - (0.9048 - 0.0952\alpha)z^{-1})}$$

$$\begin{aligned}
 (b) \quad p(z) &= (1-z^{-1})(1 - (0.9048 - 0.0952\alpha)z^{-1}) \\
 &= 1 - (0.9048 - 0.0952\alpha)z^{-1} - z^{-1} + (0.9048 - 0.0952\alpha)z^{-2} \\
 \hat{p}(z) &= z^2 - (1.9048 - 0.0952\alpha)z + (0.9048 - 0.0952\alpha)
 \end{aligned}$$

$$\text{Stable: } \textcircled{1} a_0 = 1 > |a_2|, \quad |0.9048 - 0.0952\alpha| < 1$$

$$\Rightarrow -1 < \alpha < 20.0084$$

$$\textcircled{2} \hat{p}(z=1) = 1 - 1.9048 + 0.0952\alpha + 0.9048 - 0.0952\alpha = 0$$

$$\Rightarrow \text{at least one pole at } z=1.$$

$$\begin{aligned}
 \textcircled{3} \hat{p}(z=-1) &= 1 + 1.9048 - 0.0952\alpha + 0.9048 - 0.0952\alpha \\
 &= 3.8096 - 0.1904\alpha > 0
 \end{aligned}$$

$$\Rightarrow \alpha < 20.0084. \quad \therefore -1 < \alpha < 20.0084.$$

$$(c) \quad E(z) = R(z) - C(z), \quad R(z) = \mathcal{Z}[r(t)] = \frac{0.1z^{-1}}{(1-z^{-1})^2}$$

$$C(z) = \frac{0.0952\alpha z^{-1}}{(1-z^{-1})^2 [1 - (0.9048 - 0.0952\alpha)z^{-1}]}$$

$$R(z) - C(z) = \frac{0.1z^{-1} [1 - (0.9048 - 0.0952\alpha)z^{-1}] - 0.0952\alpha z^{-1}}{(1-z^{-1})^2 [1 - (0.9048 - 0.0952\alpha)z^{-1}]}$$

$$3. (a) \begin{bmatrix} \dot{x}_1(t) \\ \dot{x}_2(t) \end{bmatrix} = \begin{bmatrix} -10 & 0 \\ 0 & -1 \end{bmatrix} \begin{bmatrix} x_1(t) \\ x_2(t) \end{bmatrix} + \begin{bmatrix} 1 \\ 2 \end{bmatrix} u(t)$$

$$\begin{cases} \dot{x}(t) = A x(t) + B u(t) \\ y(t) = C x(t) \end{cases} \quad y(t) = \begin{bmatrix} -2 & 1 \end{bmatrix} \begin{bmatrix} x_1(t) \\ x_2(t) \end{bmatrix}$$

$$(i) [sI - A]^{-1} = \begin{bmatrix} s+10 & 0 \\ 0 & s+1 \end{bmatrix}^{-1} = \frac{\begin{bmatrix} s+1 & 0 \\ 0 & s+10 \end{bmatrix}}{(s+1)(s+10)} = \begin{bmatrix} \frac{1}{s+10} & 0 \\ 0 & \frac{1}{s+1} \end{bmatrix}$$

$$\Phi(T) = \mathcal{L}^{-1}\{[sI - A]^{-1}\} = \begin{bmatrix} e^{-10T} & 0 \\ 0 & e^{-T} \end{bmatrix}$$

$$\begin{aligned} \Theta(T) &= \int_0^T \Phi(T) B d\tau = \int_0^T \begin{bmatrix} e^{-10\tau} & 0 \\ 0 & e^{-\tau} \end{bmatrix} \begin{bmatrix} 1 \\ 2 \end{bmatrix} d\tau \\ &= \int_0^T \begin{bmatrix} e^{-10\tau} \\ 2e^{-\tau} \end{bmatrix} d\tau = \begin{bmatrix} -0.1e^{-10\tau} \\ -2e^{-\tau} \end{bmatrix} \Big|_0^T = \begin{bmatrix} -0.1e^{-10T} + 0.1 \\ -2e^{-T} + 2 \end{bmatrix} \end{aligned}$$

$$\Rightarrow x(k+1) = \Phi(T) x(k) + \Theta(T) u(k)$$

$$= \begin{bmatrix} e^{-1} & 0 \\ 0 & e^{-0.1} \end{bmatrix} x(k) + \begin{bmatrix} -0.1e^{-1} + 0.1 \\ -2e^{-0.1} + 2 \end{bmatrix} u(k)$$

$$= \begin{bmatrix} 0.3679 & 0 \\ 0 & 0.9048 \end{bmatrix} x(k) + \begin{bmatrix} 0.0632 \\ 0.1903 \end{bmatrix} u(k)$$

$$y(k) = C x(k)$$

$$= \begin{bmatrix} -2 & 1 \end{bmatrix} x(k)$$

$$(ii) \det[zI - A] = \begin{vmatrix} z - 0.3679 & 0 \\ 0 & z - 0.9048 \end{vmatrix} = (z - 0.3679)(z - 0.9048)$$

$$\Rightarrow \text{poles: } p_1 = 0.3679, \quad p_2 = 0.9048.$$

$$(b) \quad C = \begin{bmatrix} C_1 & C_2 \end{bmatrix}, \quad W_0 = \begin{bmatrix} C \\ CA \end{bmatrix} = \begin{bmatrix} C_1 & C_2 \\ C_2 & C_1 \end{bmatrix}, \quad |W_0| \neq 0$$

$$\Rightarrow C_1^2 - C_2^2 \neq 0$$

$$J = \frac{1}{2} \sum_{k=0}^{\infty} (x^T(k) Q x(k) + r u^2(k))$$

$$= \frac{1}{2} \sum_{k=0}^{\infty} (q x^2(k) + 8 u^2(k)),$$

$$u^*(k) = -0.5 x(k), \quad k = 0.5, \quad r = 8, \quad A = B = 1$$

$$K = (B^T S B + r)^{-1} B^T S A$$

$$0.5 = (S + 8)^{-1} S, \quad 0.5(S + 8) = S, \quad S = 8$$

$$8 = 8 + q - 8(8 + 8)^{-1} \times 8, \quad q = 4$$

$$x(k+1) = A x(k) + B u(k) = [A - BK] x(k)$$

$$|z| - [A - BK] = |z - (1 - 0.5)| = |z - 0.5| \Rightarrow \text{pole } z = 0.5.$$

$$4. (a) \quad x(k+1) = Ax(k) + Bu(k), \quad u(k) = -Kx(k)$$

$$\Rightarrow x(k+1) = [A - BK]x(k)$$

$$= \begin{bmatrix} e^{-1} & 0 \\ 1-e^{-1} & 1 \end{bmatrix} - \begin{bmatrix} 1-e^{-1} \\ e^{-1} \end{bmatrix} \begin{bmatrix} k_1 & k_2 \end{bmatrix}$$

$$= \begin{bmatrix} e^{-1} - k_1 + k_1 e^{-1} & -k_2 + k_2 e^{-1} \\ 1 - e^{-1} - k_1 e^{-1} & 1 - k_2 e^{-1} \end{bmatrix}$$

$$\text{for deadbeat control: } K = [0 \ 1] W_c^{-1} \alpha_c(A), \quad \alpha_c(A) = A^2$$

$$W_c^{-1} = [B \ AB]^{-1} = \begin{bmatrix} 1-e^{-1} & e^{-1}-e^{-2} \\ e^{-1} & (1-e^{-1})^2 + e^{-1} \end{bmatrix}^{-1}$$

$$= \begin{bmatrix} 0.6321 & 0.2325 \\ 0.3679 & 0.7675 \end{bmatrix}^{-1} = \frac{\begin{bmatrix} 0.7675 & -0.2325 \\ -0.3679 & 0.6321 \end{bmatrix}}{0.3996}$$

$$\alpha_c(A) = A^2 = \begin{bmatrix} e^{-1} & 0 \\ 1-e^{-1} & 1 \end{bmatrix} \begin{bmatrix} e^{-1} & 0 \\ 1-e^{-1} & 1 \end{bmatrix} = \begin{bmatrix} 0.1353 & 0 \\ 0.8647 & 1 \end{bmatrix} = \begin{bmatrix} 1.9207 & -0.5818 \\ -0.9207 & 1.5818 \end{bmatrix}$$

$$\Rightarrow K = [0 \ 1] \begin{bmatrix} 1.9207 & -0.5818 \\ -0.9207 & 1.5818 \end{bmatrix} \begin{bmatrix} 0.1353 & 0 \\ 0.8647 & 1 \end{bmatrix}$$

$$= [-0.9207 \ 1.5818] \begin{bmatrix} 0.1353 & 0 \\ 0.8647 & 1 \end{bmatrix} = [1.2432 \ 1.5818]$$

$$\Rightarrow k_1 = 1.2432, \quad k_2 = 1.5818.$$

$$x(k+1) = [A - BK]x(k), \quad u(k) = -Kx(k)$$

$$(b) \quad L_0 = \alpha_0(A) W_0^{-1} \begin{bmatrix} 0 \\ 1 \end{bmatrix}, \quad \alpha_0(z) = (z - z_1)(z - z_2) = z^2 - (z_1 + z_2)z + z_1 z_2$$

$$W_0^{-1} = \begin{bmatrix} C \\ CA \end{bmatrix}^{-1} = \begin{bmatrix} 0 & 1 \\ 0.22 & 1 \end{bmatrix}^{-1} = \frac{\begin{bmatrix} 1 & -1 \\ -0.22 & 0 \end{bmatrix}}{-0.22} = \begin{bmatrix} -4.5455 & 4.5455 \\ 1 & 0 \end{bmatrix}$$

$$\alpha_0(A) = A^2 = \begin{bmatrix} 0.78 & 0 \\ 0.22 & 1 \end{bmatrix} \begin{bmatrix} 0.78 & 0 \\ 0.22 & 1 \end{bmatrix} = \begin{bmatrix} 0.6084 & 0 \\ 0.3916 & 1 \end{bmatrix}$$

$$\Rightarrow L_0 = \begin{bmatrix} 0.6084 & 0 \\ 0.3916 & 1 \end{bmatrix} \begin{bmatrix} -4.5455 & 4.5455 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} 0 \\ 1 \end{bmatrix} = \begin{bmatrix} -2.7655 & 2.7655 \\ -0.7800 & 1.7800 \end{bmatrix} \begin{bmatrix} 0 \\ 1 \end{bmatrix}$$

$$= \begin{bmatrix} 2.7655 \\ 1.7800 \end{bmatrix}$$

$$(c) \quad x_e(k+1) = x(k+1) - \bar{x}(k+1)$$

$$\bar{x}(k+1) = [A - L_0 C A] \bar{x}(k) + [B - L_0 C B] \bar{u}(k) + L_0 y(k+1)$$

$$\Rightarrow x_e(k+1) = x(k+1) - \bar{x}(k+1)$$

$$= A x(k) - [A - L_0 C A] \bar{x}(k) + B u(k) - [B - L_0 C B] \bar{u}(k) - L_0 y(k+1)$$

$$= A x_e(k) + L_0 C A \bar{x}(k) + B(u(k) - \bar{u}(k)) + L_0 C B \bar{u}(k) - L_0 C x(k+1)$$

$$5. (a) \quad p(s) = s^2 + 2\zeta\omega_n s + \omega_n^2 = s^2 + 5s + 25$$

$$\Delta = 5^2 - 4 \times 25 = -95, \quad s_{1,2} = \frac{1}{2}(-5 \pm \sqrt{\Delta}) = -\frac{5}{2} \pm j\sqrt{95}$$

$G(s)$ has $n=2$ poles, $m=0$ zero, hence:

$$z_{1,2} = e^{s_{1,2}T} \Rightarrow z_1 = e^{(-\frac{5}{2} + j\frac{\sqrt{95}}{2})T}, \quad z_2 = e^{(-\frac{5}{2} - j\frac{\sqrt{95}}{2})T}$$

$$G(z) = \frac{\alpha(z+1)}{(z-z_1)(z-z_2)} = \frac{\alpha(z+1)}{z^2 - (z_1+z_2)z + z_1z_2}$$

$$z_1+z_2 = e^{-\frac{5}{2}T} \times 2\cos(\frac{\sqrt{95}}{2}T) = 0.0821 \times 1.9711 = 0.1618$$

$$z_1z_2 = e^{-5T} = 0.00674$$

$$\Rightarrow G(z) = \frac{\alpha(z+1)}{z^2 - 0.1618z + 0.00674}$$

$$G(1) = G_a(0) = 1, \quad \frac{2\alpha}{1 - 0.1618 + 0.00674} = 1, \quad \alpha = 0.42247$$

$$\Rightarrow G(z) = 0.42247 \frac{z+1}{z^2 - 0.1618z + 0.00674}$$

$$(b) \quad C(z) = \frac{1}{G_{2AS}(z)} \frac{G_{cl}(z)}{1 - G_{cl}(z)}, \quad G_{cl}(z) = G(z)$$

$$1 - G_{cl}(z) = \frac{z^2 - 0.1618z + 0.00674 - 0.42247z - 0.42247}{z^2 - 0.1618z + 0.00674}$$

$$= \frac{z^2 - 0.58427z - 0.41573}{z^2 - 0.1618z + 0.00674}$$

$$\frac{G_{cl}(z)}{1 - G_{cl}(z)} = \frac{z+1}{z^2 - 0.58427z - 0.41573}$$

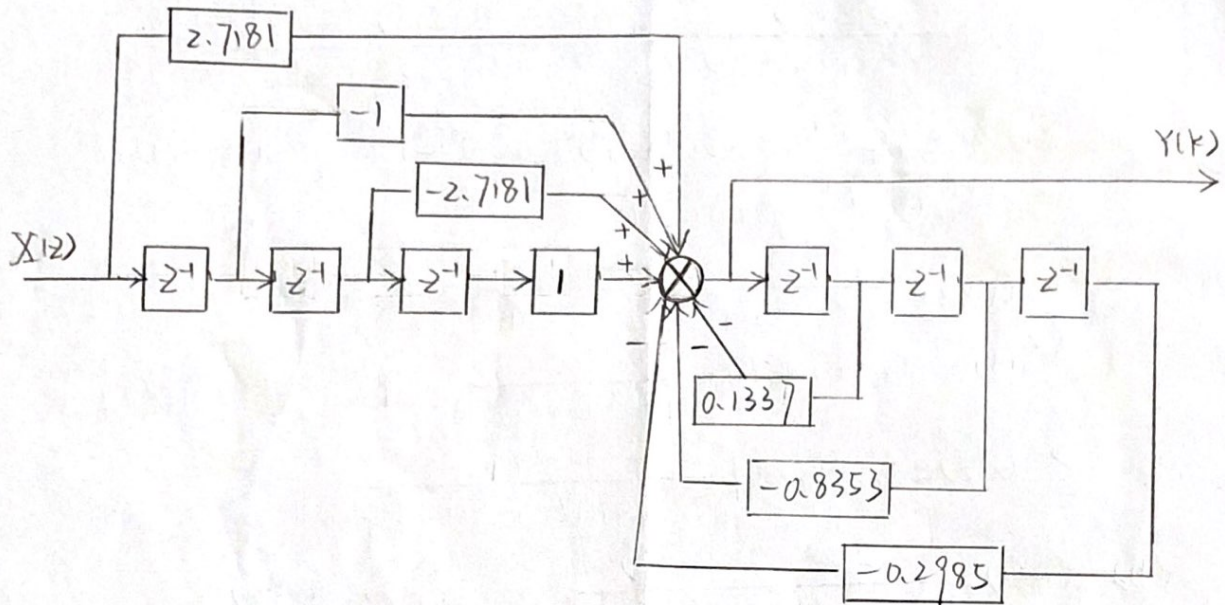
$$\Rightarrow C(z) = \frac{(z-1)(z-0.3679)}{0.3679z + 0.2642} \cdot \frac{z+1}{z^2 - 0.58427z - 0.41573}$$

$$= \frac{z^2 - 0.3679z^2 - z + 0.3679}{0.3679z^3 + 0.0492z^2 - 0.3073z - 0.1098}$$

$$= \frac{2.7181 - z^{-1} - 2.7181z^2 + z^{-3}}{1 + 0.1337z^{-1} - 0.8355z^{-2} - 0.2985z^{-3}}$$

$$(c) \frac{Y(z)}{X(z)} = \frac{2.7181 - z^{-1} - 2.7181z^{-2} + z^{-3}}{1 + 0.1337z^{-1} - 0.8353z^{-2} - 0.2985z^{-3}} = C(z)$$

$$\Rightarrow y(k) = -0.1337 y(k-1) + 0.8353 y(k-2) + 0.2985 y(k-3) + 2.7181 x(k) - x(k-1) - 2.7181 x(k-2) + x(k-3)$$



EE6203

NANYANG TECHNOLOGICAL UNIVERSITY
SEMESTER 1 EXAMINATION 2018-2019
EE6203 – COMPUTER CONTROL SYSTEMS

November/December 2018

Time Allowed: 3 hours

INSTRUCTIONS

1. This paper contains 5 questions and comprises 8 pages.
2. Answer all 5 questions.
3. All questions carry equal marks.
4. This is a closed-book examination.
5. The Transform Table is included in Appendix A on pages 7 to 8.

-
1. (a) Consider the following sampled-data signal:

$$x(kT) = \begin{cases} kT, & k \geq 0 \\ 0, & k < 0 \end{cases}$$

where T is the sampling period and k is an integer. If

$$y(kT) = \sum_{h=0}^k x(hT),$$

determine the Z-transform of $y(kT)$ and verify the Initial Value Theorem.

(6 Marks)

- (b) Suppose that $x(kT)$ is the sampled signal from $x(t)$ with sampling period $T=1$ second and its Z-transform is given as

$$X(z) = \frac{z}{(z-1)(z^2 - \sqrt{2}z + 1)}.$$

Find $x(kT)$.

(11 Marks)

Note: Question No. 1 continues on page 2

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- (c) Based on the Final Value Theorem, a student found that $x(kT)$ in Question 1(b) satisfies

$$\lim_{k \rightarrow \infty} x(kT) = \frac{1}{2} + \frac{\sqrt{2}}{2}.$$

Justify, with reasons, whether the answer from the student is correct.

(3 Marks)

2. Consider the control system shown in Figure 1.

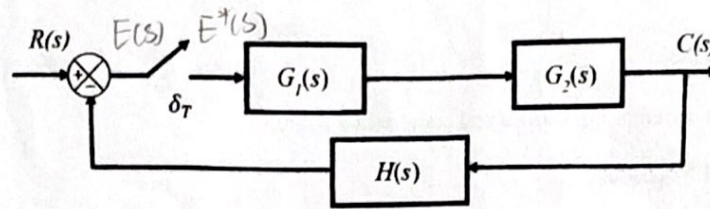


Figure 1

- (a) Suppose that

$$G_1(s) = \frac{1-e^{-Ts}}{s}, \quad G_2(s) = \frac{\alpha}{s+1} \text{ where } \alpha \text{ is a constant.}$$

If $H(s) = 1$, the sampling period T is 0.1 second and the input $r(t)$ is a unit-step function, determine the Z-transform of the system output.

(11 Marks)

- (b) Find the range of α in Question 2(a) to ensure the stability of the system.

(5 Marks)

- (c) If the reference input $r(t) = t$ for $t \geq 0$, justify whether there exists a constant α such that $\lim_{k \rightarrow \infty} |e(kT)| \leq 0.1$, where $e(kT) = r(kT) - c(kT)$.

(4 Marks)

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3. (a) The dynamics of a room temperature control system are given by the following system:

$$\begin{bmatrix} \frac{dx_1(t)}{dt} \\ \frac{dx_2(t)}{dt} \end{bmatrix} = \begin{bmatrix} -10 & 0 \\ 0 & -1 \end{bmatrix} \begin{bmatrix} x_1(t) \\ x_2(t) \end{bmatrix} + \begin{bmatrix} 1 \\ 2 \end{bmatrix} u(t)$$

$$y(t) = \begin{bmatrix} -2 & 1 \end{bmatrix} \begin{bmatrix} x_1(t) \\ x_2(t) \end{bmatrix}$$

where $x_1(t)$ and $x_2(t)$ are the states, $u(t)$ and $y(t)$ are the input and output variables, respectively. The system is sampled with a zero-order hold at a sampling period $T = 0.1$ second.

- (i) Determine a discretised state-space model for the system.
 (ii) Where are the poles of the discrete-time system located?

(10 Marks)

- (b) A discrete-time system has a state-space representation given by

$$\begin{bmatrix} x_1(k+1) \\ x_2(k+1) \end{bmatrix} = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} x_1(k) \\ x_2(k) \end{bmatrix} + \begin{bmatrix} 1 \\ 1 \end{bmatrix} u(k)$$

$$y(k) = C \begin{bmatrix} x_1(k) \\ x_2(k) \end{bmatrix}$$

Give a matrix C such that the system is observable and has a single variable output $y(k)$.

(5 Marks)

- (c) Consider a system which is described by the following state equation:

$$\mathbf{x}(k+1) = \mathbf{A}\mathbf{x}(k) + \mathbf{B}u(k)$$

and with an associated performance index given by

$$J = \frac{1}{2} \sum_{k=0}^{\infty} (\mathbf{x}^T(k) \mathbf{Q} \mathbf{x}(k) + r u^2(k))$$

Note: Question No. 3 continues on page 4

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The control law that minimises J is of the following form:

$$u^*(k) = -Kx(k)$$

where the optimal control gain is given by

$$K = (B^T S B + r)^{-1} B^T S A$$

and $S > 0$ solves the following equation:

$$S = A^T S A + Q - A^T S B (r + B^T S B)^{-1} B^T S A$$

Now, consider the following system and associated performance index:

$$x(k+1) = x(k) + u(k)$$

$$J = \frac{1}{2} \sum_{k=0}^{\infty} (qx^2(k) + 8u^2(k))$$

If the optimal control law that minimises J is

$$u^*(k) = -0.5x(k)$$

determine q . Determine also the location of the closed-loop pole.

(5 Marks)

4. (a) A discrete-time system has a state-space representation given by

$$\begin{bmatrix} x_1(k+1) \\ x_2(k+1) \end{bmatrix} = \begin{bmatrix} e^{-T} & 0 \\ 1-e^{-T} & 1 \end{bmatrix} \begin{bmatrix} x_1(k) \\ x_2(k) \end{bmatrix} + \begin{bmatrix} 1-e^{-T} \\ T-1+e^{-T} \end{bmatrix} u(k)$$

$$y(k) = \begin{bmatrix} 0 & 1 \end{bmatrix} \begin{bmatrix} x_1(k) \\ x_2(k) \end{bmatrix}$$

where $\begin{bmatrix} x_1(k) \\ x_2(k) \end{bmatrix}$, $u(k)$ and $y(k)$ are the states, input and output variables, respectively. If the sampling period T is 1 second, design the deadbeat state-feedback controller.

(7 Marks)

Note: Question No. 4 continues on page 5

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- (b) A discrete-time system has a state-space representation given by

$$\begin{bmatrix} x_1(k+1) \\ x_2(k+1) \end{bmatrix} = \begin{bmatrix} 0.78 & 0 \\ 0.22 & 1 \end{bmatrix} \begin{bmatrix} x_1(k) \\ x_2(k) \end{bmatrix} + \begin{bmatrix} 0.22 \\ 0.03 \end{bmatrix} u(k)$$

$$y(k) = \begin{bmatrix} 0 & 1 \end{bmatrix} \begin{bmatrix} x_1(k) \\ x_2(k) \end{bmatrix}$$

where $\begin{bmatrix} x_1(k) \\ x_2(k) \end{bmatrix}$, $u(k)$ and $y(k)$ are the states, input and output variables, respectively. Design a full-order prediction estimator of the following form:

$$\bar{\mathbf{x}}(k+1) = \mathbf{A}\bar{\mathbf{x}}(k) + \mathbf{B}u(k) + \mathbf{L}_o(y(k) - \mathbf{C}\bar{\mathbf{x}}(k))$$

such that the desired estimator poles are at $z_{1,2} = 0$. Verify that the error response is indeed deadbeat with the designed estimator.

(7 Marks)

- (c) A discrete-time system has a state-space representation given by

$$\mathbf{x}(k+1) = \mathbf{A}\mathbf{x}(k) + \mathbf{B}u(k)$$

$$y(k) = \mathbf{C}\mathbf{x}(k)$$

where $\mathbf{x}(k)$, $u(k)$ and $y(k)$ are the state vector, input and output variables, respectively. The following observer is employed to give an estimate of the state vector $\bar{\mathbf{x}}(k)$:

$$\bar{\mathbf{x}}(k) = \mathbf{A}\bar{\mathbf{x}}(k-1) + \mathbf{B}u(k-1) + \mathbf{L}_o\{y(k) - \mathbf{C}(\mathbf{A}\bar{\mathbf{x}}(k-1) + \mathbf{B}u(k-1))\}$$

If the error vector is defined as

$$\mathbf{x}_e(k) = \mathbf{x}(k) - \bar{\mathbf{x}}(k)$$

obtain an equation for the error dynamics of the observer in the following form:

$$\mathbf{x}_e(k+1) = \mathbf{A}_e\mathbf{x}_e(k)$$

i.e. obtain an expression for $\mathbf{A}_e$ in terms of $\mathbf{A}$, $\mathbf{B}$, $\mathbf{C}$ and $\mathbf{L}_o$, where applicable.

(6 Marks)

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5. (a) Consider the following second-order analog filter:

$$G(s) = \frac{\omega_n^2}{s^2 + 2\zeta\omega_n s + \omega_n^2}$$

$$G(z) = \frac{\alpha(z+1)}{(z-z_1)(z-z_2)}$$

where $\omega_n = 5$ rad/s and $\zeta = 0.5$. Find its pole-zero matched strictly proper digital filter approximation with the same DC gain and a sampling period of 1 second.

(8 Marks)

- (b) Consider the closed-loop system shown in Figure 2.

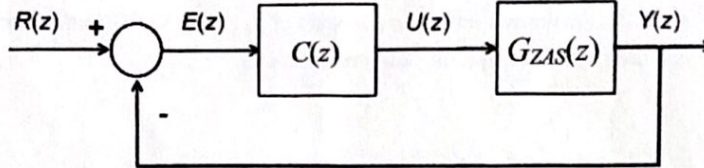


Figure 2

The desired closed-loop transfer function $G_{cl}(z)$ has the same specifications as those of the filter in Question 5(a). With a sampling period of 1 second, $G_{ZAS}(z)$ is given as

$$G_{ZAS}(z) = \frac{0.3679z + 0.2642}{(z-1)(z-0.3679)}$$

Design the controller $C(z)$.

(8 Marks)

- (c) Implement the controller $C(z)$ obtained in Question 5(b) with the direct programming approach and show the relevant block diagram.

(4 Marks)

Appendix A

Properties and Table of Z-Transform

Discrete function	z Transform
$x(k+4)$	$z^4 X(z) - z^4 x(0) - z^3 x(1) - z^2 x(2) - zx(3)$
$x(k+3)$	$z^3 X(z) - z^3 x(0) - z^2 x(1) - zx(2)$
$x(k+2)$	$z^2 X(z) - z^2 x(0) - zx(1)$
$x(k+1)$	$zX(z) - zx(0)$
$x(k)$	$X(z)$
$x(k-1)$	$z^{-1} X(z)$
$x(k-2)$	$z^{-2} X(z)$
$x(k-3)$	$z^{-3} X(z)$
$x(k-4)$	$z^{-4} X(z)$

	$X(s)$	$x(t)$	$x(kT)$ or $x(k)$	$X(z)$
1.	—	—	Kronecker delta $\delta_0(k)$ 1, $k=0$ 0, $k \neq 0$	1
2.	—	—	$\delta_0(n-k)$ 1, $n=k$ 0, $n \neq k$	z^{-k}
3.	$\frac{1}{s}$	$1(t)$	$1(k)$	$\frac{1}{1-z^{-1}}$
4.	$\frac{1}{s+a}$	e^{-at}	e^{-akt}	$\frac{1}{1-e^{-aT}z^{-1}}$
5.	$\frac{1}{s^2}$	t	kT	$\frac{Tz^{-1}}{(1-z^{-1})^2}$
6.	$\frac{2}{s^3}$	t^2	$(kT)^2$	$\frac{T^2 z^{-1}(1+z^{-1})}{(1-z^{-1})^3}$
7.	$\frac{6}{s^4}$	t^3	$(kT)^3$	$\frac{T^3 z^{-1}(1+4z^{-1}+z^{-2})}{(1-z^{-1})^4}$
8.	$\frac{a}{s(s+a)}$	$1-e^{-at}$	$1-e^{-akt}$	$\frac{(1-e^{-aT})z^{-1}}{(1-z^{-1})(1-e^{-aT}z^{-1})}$
9.	$\frac{b-a}{(s+a)(s+b)}$	$e^{-at} - e^{-bt}$	$e^{-akt} - e^{-bkt}$	$\frac{(e^{-aT} - e^{-bT})z^{-1}}{(1-e^{-aT}z^{-1})(1-e^{-bT}z^{-1})}$
10.	$\frac{1}{(s+a)^2}$	te^{-at}	kTe^{-akt}	$\frac{Tze^{-aT}z^{-1}}{(1-e^{-aT}z^{-1})^2}$
11.	$\frac{s}{(s+a)^2}$	$(1-at)e^{-at}$	$(1-akt)e^{-akt}$	$\frac{1-(1+aT)e^{-aT}z^{-1}}{(1-e^{-aT}z^{-1})^2}$

Note: Transform Table continues on page 8.

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	$X(s)$	$x(t)$	$x(kT)$ or $x(k)$	$X(z)$
12.	$\frac{2}{(s+a)^3}$	$t^2 e^{-at}$	$(kT)^2 e^{-akT}$	$\frac{T^2 e^{-aT} (1 + e^{-aT} z^{-1}) z^{-1}}{(1 - e^{-aT} z^{-1})^3}$
13.	$\frac{a^2}{s^2(s+a)}$	$at - 1 + e^{-at}$	$akT - 1 + e^{-akT}$	$\frac{[(aT - 1 + e^{-aT}) + (1 - e^{-aT} - aT e^{-aT}) z^{-1}]}{(1 - z^{-1})^2 (1 - e^{-aT} z^{-1})} z^{-1}$
14.	$\frac{\omega}{s^2 + \omega^2}$	$\sin \omega t$	$\sin \omega kT$	$\frac{z^{-1} \sin \omega T}{1 - 2z^{-1} \cos \omega T + z^{-2}}$
15.	$\frac{s}{s^2 + \omega^2}$	$\cos \omega t$	$\cos \omega kT$	$\frac{1 - z^{-1} \cos \omega T}{1 - 2z^{-1} \cos \omega T + z^{-2}}$
16.	$\frac{\omega}{(s+a)^2 + \omega^2}$	$e^{-at} \sin \omega t$	$e^{-akT} \sin \omega kT$	$\frac{e^{-aT} z^{-1} \sin \omega T}{1 - 2e^{-aT} z^{-1} \cos \omega T + e^{-2aT} z^{-2}}$
17.	$\frac{s+a}{(s+a)^2 + \omega^2}$	$e^{-at} \cos \omega t$	$e^{-akT} \cos \omega kT$	$\frac{1 - e^{-aT} z^{-1} \cos \omega T}{1 - 2e^{-aT} z^{-1} \cos \omega T + e^{-2aT} z^{-2}}$
18.			a^k	$\frac{1}{1 - az^{-1}}$
19.			a^{k-1} $k = 1, 2, 3, \dots$	$\frac{z^{-1}}{1 - az^{-1}}$
20.			ka^{k-1}	$\frac{z^{-1}}{(1 - az^{-1})^2}$
21.			$k^2 a^{k-1}$	$\frac{z^{-1}(1 + az^{-1})}{(1 - az^{-1})^3}$
22.			$k^3 a^{k-1}$	$\frac{z^{-1}(1 + 4az^{-1} + a^2 z^{-2})}{(1 - az^{-1})^4}$
23.			$k^4 a^{k-1}$	$\frac{z^{-1}(1 + 11az^{-1} + 11a^2 z^{-2} + a^3 z^{-3})}{(1 - az^{-1})^5}$
24.			$a^k \cos k\pi$	$\frac{1}{1 + az^{-1}}$
25.			$\frac{k(k-1)}{2!}$	$\frac{z^{-2}}{(1 - z^{-1})^3}$
26.		$\frac{k(k-1) \dots (k-m+2)}{(m-1)!}$		$\frac{z^{-m+1}}{(1 - z^{-1})^m}$
27.			$\frac{k(k-1)}{2!} a^{k-2}$	$\frac{z^{-2}}{(1 - az^{-1})^3}$
28.		$\frac{k(k-1) \dots (k-m+2)}{(m-1)!} a^{k-m+1}$		$\frac{z^{-m+1}}{(1 - az^{-1})^m}$

$x(t) = 0$, for $t < 0$.

$x(kT) = x(k) = 0$, for $k < 0$.

Unless otherwise noted, $k = 0, 1, 2, 3, \dots$

END OF PAPER